Engram: Personality-Parameterized Schema Memory for NPC Cognitive Diversity

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Abstract

Current NPC memory systems do not encode events based on character personality, limiting cognitive diversity to surface-level dialogue variation, leading to similar or personality-breaking dialogues. We present a system that parameterizes OCEAN (Big Five) personality trait scores with a Prolog-based memory encoding rule enabling NPCs to form structurally different memories from identical experiences. Preliminary evaluation across contrasting personality configurations shows that the resulting memory representations diverge in trait-predicted directions: high-Neuroticism profiles store proportionally more threat-tagged memories, high-Extraversion profiles amplify social event encoding, and low-Openness profiles resist belief revision. This architecture enables emergent cognitive diversity without scripting, grounding NPC memory processing in an established psychological framework.

Keywords

NPC memory, personality modeling, OCEAN Big Five, schema memory, game AI, Prolog, cognitive architecture

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1 Introduction

Large language model (LLM) driven non-player characters (NPCs) are increasingly capable of generating naturalistic and contextually appropriate dialogue, but the memory systems underlying these characters remain personality-agnostic. In many current architectures, experiences are treated as neutral storage and are recorded in a largely uniform form. These systems rely on a retrieval mechanism that is only applied at the moment of response generation. Systems such as MemoryBank [11] and the generative agent architecture [6] store and retrieve memories using uniform processes regardless of which character is remembering. Personality, when present, is applied only at the output layer: the same memory is retrieved and then filtered through a character prompt to produce personality-flavored dialogue. As a result, NPCs who witness the same event will form identical internal representations of it, differing only in how they talk about it afterward.

This design brings a limitation that personality becomes a surface level filter rather than a cognitive mechanism. This is psychologically implausible. Research in personality psychology has established that personality traits systematically influence memory encoding, not just expression. Individuals high in Neuroticism encode threatening stimuli more strongly; those high in Extraversion preferentially attend to social information. The Big Five (OCEAN) model provides a well-validated framework for these individual differences [9]. In the believable agents literature, Loyall [3] identified personality consistency as a foundational requirement for character believability, arguing that personality must infuse everything a character does, including how they think and perceive, not merely how they speak.

We present a system that addresses this gap by parameterizing memory encoding rules with OCEAN personality trait scores. Events are first tagged by an LLM with structured semantic labels (threat level, social type, emotional valence, goal relevance, novelty). These tagged events are then processed by Prolog rules whose thresholds and weights are set by the NPC's personality vector. The result is that two NPCs experiencing the same event produce structurally different memory representations. Unlike prior systems/architecture, personality is integrated at the point of memory formation rather than applied later on at generation time. Our contributions are: (1) an architecture bridging LLM semantic tagging with personality-parameterized Prolog encoding rules, (2) a preliminary evaluation demonstrating trait-consistent memory divergence across personality profiles, and (3) a working prototype of the system.[1]

2 Related Work

Believable agents and personality consistency. The CMU Oz Project established the foundational requirements for believable characters in interactive systems. Bates [1] argued that emotion is central to the illusion of life in agents, while Loyall [3] formalized a set of believability requirements including personality, self-motivation, and change over time, emphasizing that breaking consistency, even momentarily, destroys suspension of disbelief. Mateas and Stern [4] encountered this tension directly in Façade, where maintaining character consistency under free-form player input proved to be the dominant engineering challenge. More recent work on identity drift in LLM agents [2] has shown that persona consistency degrades measurably after 8–12 dialogue turns, confirming that the consistency problem identified by the believable agents community persists in modern architectures. These works establish that personality consistency is essential but focus primarily on behavioral output rather than internal memory processes.

Memory in AI agents. Structured memory representations have a long history in AI, from Schank and Abelson's scripts [8] to Minsky's frames [5], both of which model how prior knowledge

[1] A working prototype is available upon request.

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shapes the encoding of new experience. Piaget's [7] concepts of assimilation (fitting new information into existing schema) and accommodation (restructuring schema when information does not fit) provide a cognitive basis for how memory structures evolve. In the LLM agent space, Park et al. [6] introduced a memory stream architecture for generative agents with retrieval based on recency, importance, and relevance. Zhong et al. [11] proposed MemoryBank, incorporating an Ebbinghaus forgetting curve and personality modeling for long-term companions. While both these systems improve memory persistence and retrieval significantly, they would assume a uniform encoding process across agents, every agent stores every event the same way, with personality affecting only downstream retrieval or generation. The ranking, retrieval and summarization is influenced by personality but not the structure of stored memories themselves. Our approach differs by limiting the encoding stage, allowing the same event to produce fundamentally different memory traces across agents, rather than barely influencing memories that are retrieved later.

OCEAN personality in computational agents. The Big Five (OCEAN) model is the most empirically validated dimensional framework in personality psychology. Sorokovikova et al. [9] demonstrated that LLMs exhibit measurable and moderately stable OCEAN trait profiles that can be shaped through prompting. Völkel et al. [10] found that conversational agents exhibit additional personality dimensions beyond OCEAN, suggesting the framework has limitations for non-human agents. Despite these limitations, OCEAN remains practical due to its continuous representation of a persona and strong empirical grounding making it suitable for setting frameworks. In contrast to prior systems that use OCEAN to guide dialogue style or response tone primarily, we will use it as a control mechanism for memory encoding and belief updating, enabling deeper cognitive diversity which is more unique rather than surface level variations.

3 System Architecture

The system processes NPC experiences through a multi-stage pipeline: threat assessment with transientdynamic personality adjustment, personality-weighted memory retrieval, mode-dependent response generation, memory contradiction checking via Prolog, and tiered memory consolidation. Each NPC maintains an OCEAN personality vector with trait values on a continuous scale, which is the basis for all memory procedures, including rolling session memory, a long-term summary store, and a set of key memories encoded as Prolog facts.

3.1 Threat Assessment

When a player message arrives, the system first performs a threat assessment. This is a quick litmus test representing our nervous system, i.e. Our memories go to Amygdala (A part of our brain assessing danger in the situation, responsible for us feeling the fear in our day-to-day lives) initially. This evaluation runs against all stored memories using cosine similarity via a simple retrieval-augmented approach, resulting in separate execution traces.

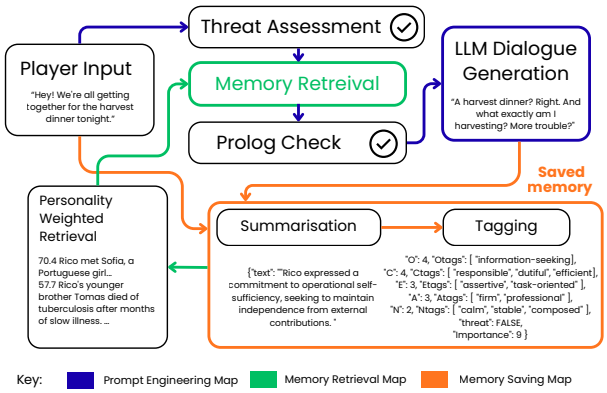

Figure 1: System architecture showing the full memory pipeline from player input through threat assessment, personality-weighted retrieval, Prolog contradiction checking, LLM dialogue generation, and tiered memory consolidation with OCEAN-biased tagging and summarization.

3.2 Personality-Weighted Retrieval

The event is then embedded using a vector embedding model and compared with all stored memories via cosine similarity, through a simple retrieval augmented system (RAG). Rather than retrieving memories by semantic similarity alone, the system applies a personality-weighted scoring formula:

$$\text{score} = (\text{RAG}_{\text{similarity score}} \times 2) \times \sum_{t \in \text{OCEAN}} \frac{t_{\text{mem}}}{t_{\text{agent}}} \times \text{importance} \quad (1)$$

where t_{mem} is the trait score tagged on the memory and t_{agent} is the NPC's current trait value. Importance is scored using LLM as judge on how important the current event is to them. This allows an NPC with a high Neuroticism score to have threat-tagged memories disproportionately higher, while an NPC with high Extraversion will prefer the retrieval of more socially-tagged memories. This is a computational low-level representation of the mental schema our brains follow when a memory is triggered. The top five memories scoring above a threshold (score ≥ 15) are selected for context in every prompt to create and save memories.

3.3 Response Mode Selection

The retrieval score determines which of three cognitive modes the NPC operates in, modeling different levels of cognitive engagement:

- **Standard Response mode** (score ≥ 15, no threat): When the NPC has strong relevant memories, and no immediate threat is detected, it proceeds to a standard response mode. The prompt used for standard responses are, the top five retrieved memories using personality weighed retrieval and a personality-biased summary are injected into the response prompt, producing contextually grounded, character-consistent dialogue.
- **Fight/flight mode** (threat detected): If the LLM evaluates the situation as a threat with context to past experiences and

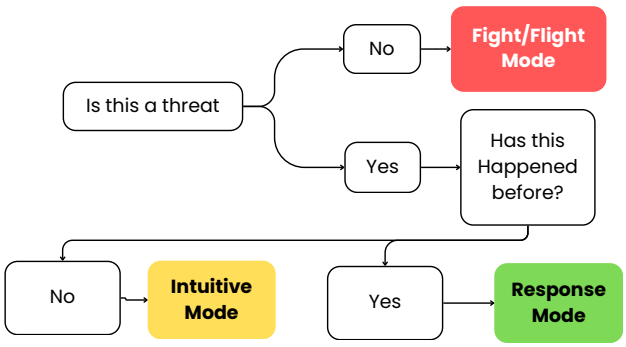

Figure 2: Response mode selection based on threat assessment and memory retrieval score.

current dialog, it produces a Fight/Flight response which produces OCEAN deltas, small signed adjustments to the NPC's personality vector (e.g., `O: + 1, N: + 2` for a threatening encounter). These deltas are applied to the NPC's current personality scores, allowing the NPC to stimulate a temporary personality shift. This is paired with a decay counter, so the personality shift is temporary: the NPC's traits gradually return toward their baseline over subsequent interactions. This mechanism models how acute experiences temporarily alter cognitive processing without permanently rewriting character identity, analogous to Piaget's [7] distinction between transient accommodation and stable schema structure.

- **Instinct mode** (score < 15, no threat): When the NPC has neither relevant memories nor a perceived threat, the context becomes more about the character than the situation, forcing the LLM to rely on intuition more than context. Retrieval uses tag-based matching against key memories only.

3.4 Prolog Contradiction Checking

Before the response is finalized and utilised, the system checks for contradictions between the new information and the NPC's existing prolog fact knowledge base, in normal facts like relationships between different characters, or general day-to-day facts. The LLM identifies potential contradictions in natural language, and SWI-Prolog (accessed via pyswip) formally verifies whether the contradiction exists in the NPC's stored facts. If a contradiction is detected, the system is sent the same prompt with the contradiction flagged, preventing the NPC from hallucinating beliefs inconsistent with its established knowledge, previous conversations, and backstories.

When general facts of the stories change, the engine handles it by removing the fact from the database for the session. After the session, based on the NPC's Openness scores, the contradiction is resolved: high-O NPCs accommodate the new information and update their beliefs more willingly, while low-O NPCs reject it and maintain existing facts.

3.5 Memory Consolidation

The system maintains a three-tier memory hierarchy:

- **Session memory**: A rolling window of the most recent seven messages, providing immediate conversational context. This follows George Miller's theory of "The Magical Number Seven, Plus or Minus Two" (1956), which states that the average human short-term memory can hold approximately 7 ± 2(5 to 9) "chunks" of information at once.
- **Long-term summaries**: Every five messages evicted from session memory, the system generates an OCEAN-biased two-line summary and stores it in the long-term memory store. These summaries are weighted by the NPC's personality and tagged based on underlying labels of OCEAN personality traits, so the same conversation produces different long-term traces for different personalities.
- **Key memories**: Memories scoring in the top 25% by the personality-weighted formula are promoted to a key memory store. New key events additionally produce Prolog facts using LLM, validated by SWI-Prolog, and appended to the NPC's fact base for use in future contradiction checking.

This tiered structure reinforces personality influences by not only how memories are initially encoded but also which memories persist, how they are summarized, and which are core to the character's knowledge.

4 Preliminary Evaluation

To evaluate whether personality parameterization produces meaningful, diverse memory graphs, we ran a fixed sequence of events through multiple distinct NPC personality profiles and compared the resulting memory representations.

4.1 Setup

We defined three personality profiles representing contrasting trait configurations:

(1) *Paranoid Guard* (high-N, low-A): ocean(0.2, 0.5, 0.3, 0.2, 0.9)
(2) *Friendly Merchant* (high-E, high-A): ocean(0.5, 0.5, 0.9, 0.8, 0.2)
(3) *Rigid Clerk* (low-O, high-C): ocean(0.1, 0.9, 0.3, 0.5, 0.4)

Each profile processed an identical sequence of six events progressively, including social encounters, threatening situations, goal-relevant tasks, and novel occurrences.

4.2 Events Discussed

All three personality profiles were prompted through six different events, consisting of an average of 13-14 dialogue exchange. Here are the different events:

(1) *Physical Threat:* A stranger demanding a storeroom key
(2) *Social opportunity:* An invitation to a community harvest dinner
(3) *Contradictory information:* Evidence a trusted Mayor is embezzling
(4) *Cooperative response:* A child asking for help for a sick parent
(5) *Goal-relevant task:* An early shipment at the docks
(6) *Ambiguous novel stimulus:* A strange animal in town square

4.3 Results

We compared the resulting memory graphs across distinct profiles using three metrics: the proportion of events stored per tag category, average importance weights per tag type, and the number of belief revision (prolog contradictions) operations triggered. Table 1 summarizes the results.

Table 1: Memory encoding differences across personality profiles for identical event sequences (6 events each).

Metric	Guard	Merchant	Clerk
Avg. Neuroticism score	5.0	1.2	2.0
Avg. Agreeableness score	1.3	4.8	2.5
Avg. Conscientiousness	4.7	4.0	4.5
Threat events (%)	100	17	67
Avg. importance	9.7	7.8	7.8

4.4 Threat Triggers

The paranoid guard flagged 100% of events as threats, including a dinner invitation and a child asking for help, while the friendly merchant flagged only 17%. Interestingly, the clerk, only flagged events that were a threat to his work, and would reduce hours or money that they earn. The guard was mostly mistrusting, keeping away from the crowd and very self dependent, triggering the threat response every time someone tries to approach him.

4.5 Trait Alignment

The trait scores tracked predictions: N (Neuroticism) averaged 5.0 for the guard vs 1.2 for the merchant. A (Agreeableness) averaged 4.8 for the merchant vs 1.3 for the guard. An interesting example here would be, the merchant approached the "strange creature" with curiosity and interest, while the guard mostly regarded the animal as a problem to solve.

4.6 Importance Valence

Then the importance scores — the guard averaged 9.7, treating everything as critical, while the merchant and clerk both averaged 7.8. The guard also got stuck on things in conversations such as after the physical threat, he became even more defensive of their work and would refuse to talk about it. Similarly, when the clerk was told about the mayor, he kept asking for details and insights from different people.

5 Discussion

The results show that personality parameterization produces measurable, trait-consistent divergence across a full memory life cycle. The same events produced structurally different memory representations — triggering different memories, different threat classifications, different importance scores, different tag distributions, and, most importantly, different narrative framing. These differences emerged from the OCEAN-parameterized encoding rules, utilized in memory parsing, tagging, retrieval, saving, and summarization rather than from different prompts or scripted behavior, suggesting that personality-driven schema memory is a viable path toward cognitive diversity in NPCs.

This matters because a lot of current LLM-based NPCs achieve variation only at the dialogue surface despite their underlying cognitive processes being consistent and uniform. In contrast, Engram changes from being a stylistic modifier to a property of memory formation. This allows NPCs to change how they speak but also what they notice, what they remember, and what they prioritize. For game design, this has significant implications for replayability and emergent storytelling.

5.1 Limitations

The following limitations may constrain the current work. The tagging layer uses LLM as judge for logic, which is a potential point of failure: if the tagger misclassifies an event's threat level or social type, the memory is embedded in that specific tag and can't be changed.Secondly, we have not yet conducted a player-facing evaluation to determine whether memory divergence translates to perceived personality differences in NPC behavior. Finally, the current evaluation is conducted on a small, controlled set of scripted events, which may not fully capture the complexity and variability of real player interactions in open-ended environments.

5.2 Future work

Future work will address these gaps by working on a model that is more efficient at tagging and conducting a player study measuring whether personality-parameterized memory produces NPCs that are perceived as having more distinct and consistent personalities. We also plan to extend the dynamic OCEAN mechanism to support permanent trait shifts, allowing NPC personalities to evolve through experience. Lastly, we also plan on developing our architecture as close to psychological models of memory and event processing as possible.

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